Vipul Jain

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Professional Summary

Innovative and results-driven Data Scientist with 4+ years of experience in AI-powered software development and data-driven solutions in the automotive domain. Skilled in developing end-to-end ML and DL models, from data preprocessing to model deployment. Proficient in Python, SQL, TensorFlow, Power BI, and AWS, with a deep understanding of scalable ML pipelines and MLOps practices. Adept at cross-functional collaboration, model performance evaluation, and solving complex real-world problems using data.

Experience

[11th Sept 2023] – [Current]

Engineer – Daimler Truck Innovation Centre India

- Engineered a smart driver monitoring system that integrates real-time physiological data analysis, contributing to a projected 30% reduction in on-road incidents caused by sudden driver health issues during pilot testing.
- Developed an AI assistant using GPT-4 to analyze OBD-II fault codes and sensor data, generating human-like diagnostic reports with repair recommendations. Integrated LLM-based reasoning and structured data inputs to accelerate root-cause identification.
- Invented a first-of-its-kind vehicle-adjacent activity monitoring system aimed at enhancing driver safety and reducing property loss, with scalability potential across 10,000+ public transport units, patent in progress.
- Engineered a data-driven model to forecast ball screw degradation in CNC machines, applying time-series analysis and anomaly detection; earned Top 2.4% placement among 250 teams in a predictive maintenance competition.
- Developed a Central Gateway Efficiency and effectiveness analysis using Python Scripting eradicating manual effort.
- Developed body control features for Mercedes Benz trucks.

[21st Jan 2021] – [8th sept. 2023]

Associate Software Engineer - Bosch Global Software Technologies

- Engineered a high-accuracy AI model using LSTM networks to predict neutral gear states, achieving 95%+ accuracy in validation tests and eliminating the need for dedicated gear position sensors.

- Developed a tool to automate the Post-delivery checklist filling using python and excel reducing manual effort by 90%.
- Developed Engine specific critical safety features using MATLAB.

Projects

1. Customer Churn Analysis

- **Objective**: Analysed customer behaviour to identify patterns leading to churn for a telecommunications company.
- **Technologies**: Python, Matplotlib, Seaborn, Decision trees
- **Learning**: Learnt EDA, Visualization and effective model training

2. Real-time Sentiment Analysis

- **Objective**: Analysed social media sentiment using natural language processing.
- **Technologies**: NLTK, TensorFlow
- **Outcome**: Delivered a 90% accuracy on sentiment classification.

3. Diabetic prediction web application

- **Objective:** Trained a model to detect if the person is diabetic or not.
- **Technologies:** Python, AWS, Flask
- **Outcome:** Was able to detect if the person was diabetic or not in real time by taking real time vital inputs from user with 94% accuracy.

4. System volume control using fingers

- **Objective:** Control system volume of laptop using fingers
- **Technologies:** Python, Mediapipe, OpenCV
- **Outcome:** system volume could be controlled by pinching the index finger and thumb closer and farther.

5. Sales Data Analysis report using PowerBI

- **Objective:** Analyse Sales Data, draw meaningful insights and prepare a Dashboard in PowerBI
- **Technologies:** Excel, PowerBI,
- **Outcome:** Prepared a PowerBI dashboard highlighting key insights from data.

Certifications

- Data Science & AI Program with Domain Specialization

Education

Bachelor of Engineering in Electronics and Communication Engineering
VTU,

Bangalore Institute of Technology, Bengaluru, Karnataka
Graduation Year:2020

GPA: 7.22

Technical Skills

- **Programming Languages**: Python, SQL
- **Machine Learning**: Regression, Classification, Clustering, Decision Trees, Random Forests, XGBoost
- **Deployment on Azure/AWS**
- **Deep Learning**: Neural Networks, CNNs, RNNs, TensorFlow, LSTMs
- **Data Visualization**: Power BI, Matplotlib, Seaborn
- **GenAI**: OpenAI GPT, LangChain, RAG, semantic/graphical search
- **Data Manipulation**: Pandas, NumPy, Excel
- **Databases**: MySQL
- **Cloud Platforms**: AWS (EC2), Azure ML
- **Version Control**: Git, GitHub, Source Tree, Bitbucket
- **Soft Skills**: Problem-solving, Communication, Teamwork

Additional Information

- Availability: Immediately
- Languages: English (Fluent), Hindi (Fluent)
- Hobbies: Making AI projects, Learning new technologies, Chess, Music